MANUFACTURING PROCEDURE 80/412
GE CXK4 (LCC) Liquid Helium Transfer

1st Issue: 12/30/2011 **REVISION: 0** **This Issue: 12/30/2011**

1.0 OBJECTIVE

To establish a procedure for performing a liquid helium transfer on a GE CXK4 (LCC) Magnet.

2.0 SCOPE

This procedure applies to GE LCC magnets manufactured by General Electric. It is only to be carried out by OiService employees qualified and trained to perform this procedure. Any troubleshooting that is normally performed and the detail of the helium transfer procedure is handled outside of this MSI.

3.0 RESPONSIBILITY

It is the responsibility of the Field Service Technicians performing the work to ensure that liquid helium transfer is carried out in accordance with this procedure.

4.0 TOOLS AND PARTS REQUIRED

4.1 Required PPE: Cryogen Gloves, Face Shield, and Safety Glasses.
4.2 Portable Oxygen Meter
4.3 Dewar End Syphon(s)
4.4 12’ and/or 15’ Liquid Transfer Line
4.5 14.75” Magnet Stinger w/ Brass Cap and O-Ring
4.6 Non-Ferrous Slip Joint Pliers; Non- Ferrous Adjustable Wrench
4.7 110v Heat Gun / Extension Cord
4.8 Low Pressure Regulator
4.9 Snoop Leak Detector

5.0 SAFETY

5.1 Protective clothing, cryogenic gloves, and safety glasses with face shield must be worn during the following procedure.
5.2 Trained personnel only should perform the following procedure.
5.3 Read all safety cautions and warnings.

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5.4 Only non-magnetic equipment such as dewars, non-magnetic gas cylinders and tools may be taken into the magnet room

5.5 Open entrance and exit doors during transfer. Ensure that no unauthorized person is allowed into the imaging suite while the doors are open.

5.6 Use appropriate warning signs and cordon off area to limit access.

5.7 Never leave a helium fill unattended.

6.0 PROCEDURE

Warning: One trained OiService FST is required to perform this service. For personal safety and to minimize the risk of equipment damage, this process is to be executed only by OiService personnel who are trained to perform this procedure. Failure to follow this requirement could result in serious personal injury.

Warning: Don all protective clothing such as cryogenic gloves, safety glasses and face shield and personal oxygen monitor before continuing with procedure.

Warning: Only non-magnetic equipment such as dewars, non-magnetic gas cylinders, and tools may be taken into the magnet room.

6.1 Observe vessel pressure, normal pressure 3.9 – 4.1 psi

6.2 Record liquid helium level and magnet pressure on Service Report.

6.3 Turn the key on the Magnet Monitor clockwise to the “Fill” position; this will temporarily disable the magnet pressure builder as shown in figure 1.

Figure 1

Warning:	**The magnet pressure must be reduced to < 2.0 psi for personal safety and to insure a good liquid transfer.**

6.4 Vent magnet down to < 2.0 psi using the Vent Valve Handle (V2) as shown in Figure 2.

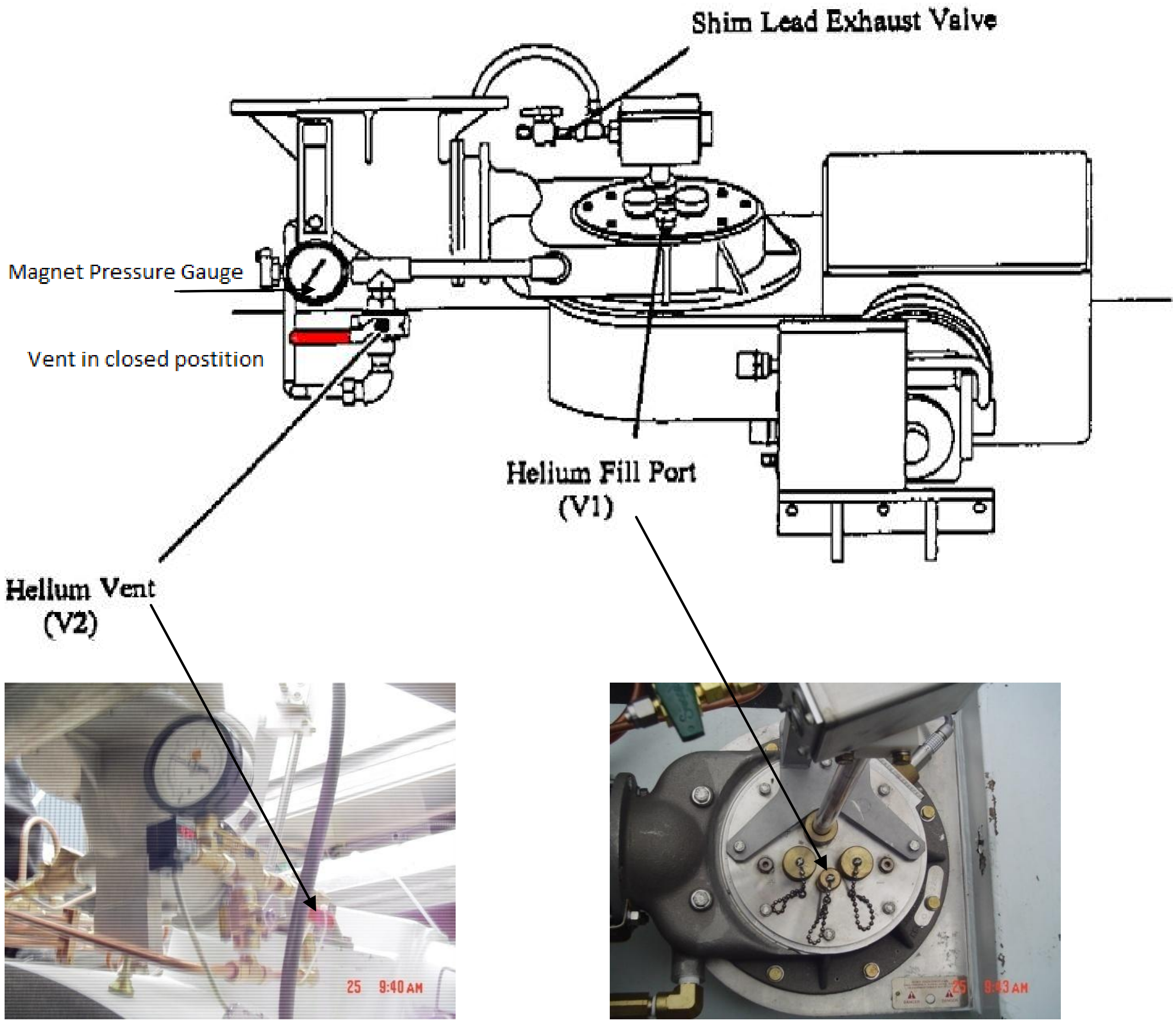

Figure 2

Figure 3

6.5 While the magnet is depressurizing, prepare the helium dewar by inserting the dewar end siphon into the dewar as shown in figure 4.

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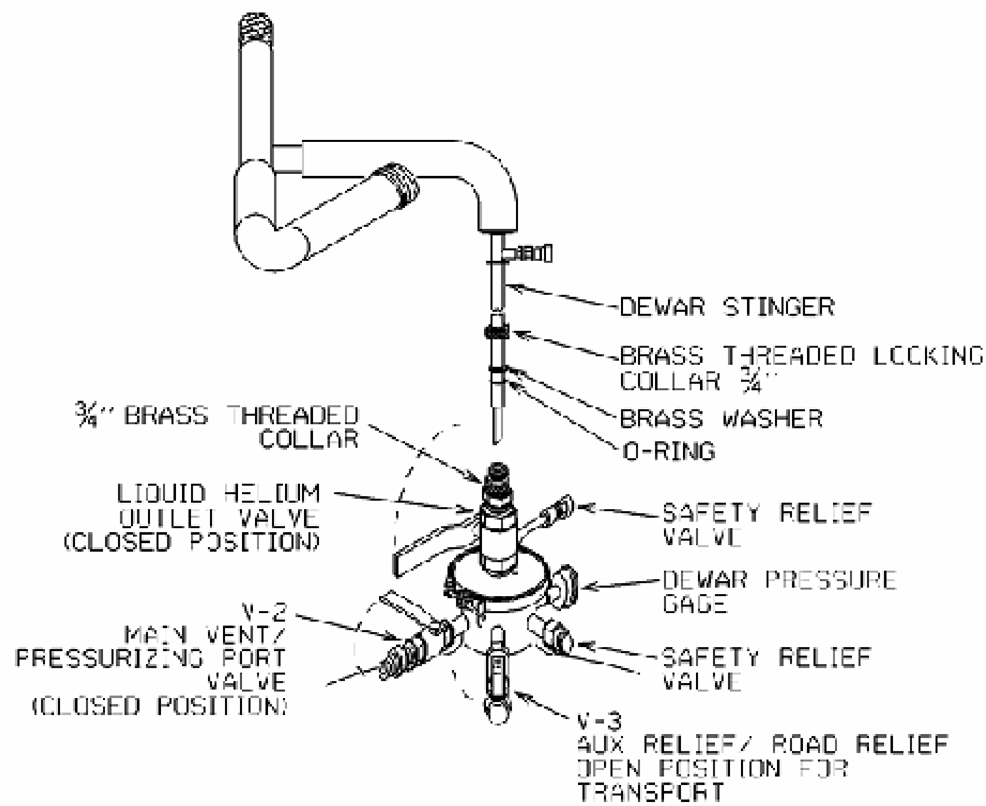

Figure 4

6.6 Connect the helium gas supply to the vent side of helium dewar or connect the 110 volt electrical source to the pressure builder to pressurize the dewar.

6.7 Connect siphon extension to the dewar end siphon as shown in figure 5.

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Figure 5

6.8 Attach the 14.75" magnet end siphon to the extension and fit the brass adaptor with ½" washer and o-ring. Assemble brass fill cap, o-ring, and washer as shown in figure 6.

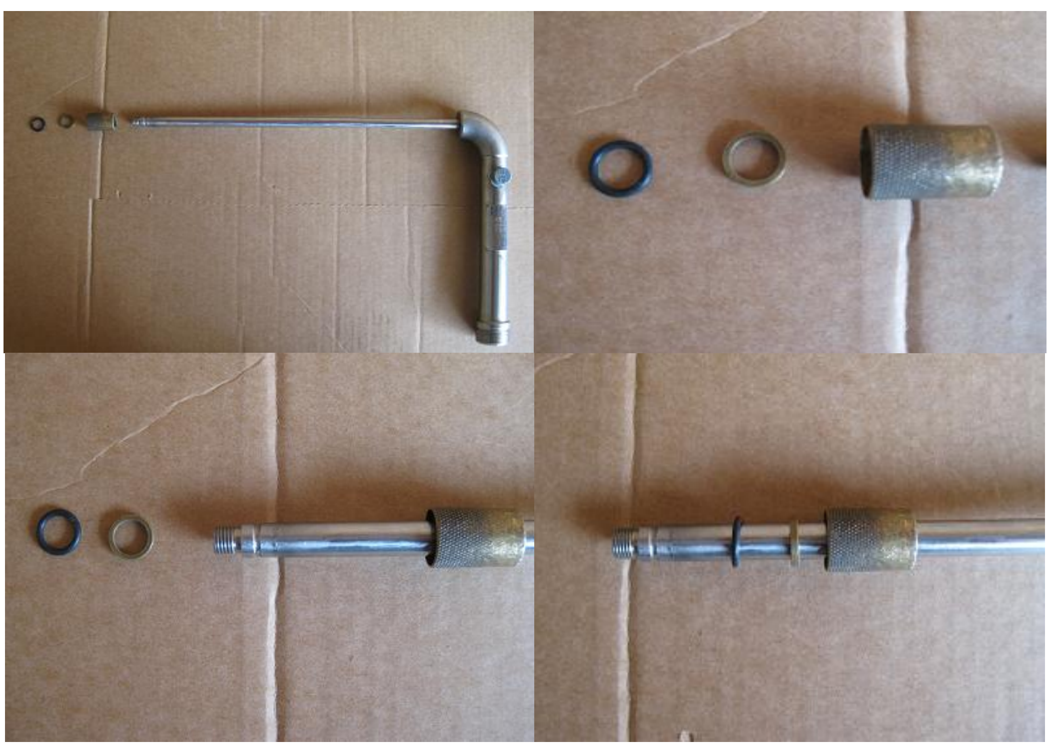

Figure 6

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Caution: Dewar pressure <u>must</u> be higher than magnet pressure. If magnet pressure is higher than dewar pressure, the magnet may quench when starting transfer!

6.9 Ensure that dewar pressure is at 4.0 psi and magnet pressure is < 2.0 psi

6.10 Once the magnet pressure has dropped below 2.0 psi, partially open the dewar end siphon valve and purge the transfer line, the extension, and the 14.75” magnet end siphon until a plume is visible as shown in Figure 7.

Figure 7

6.11 With a soft, steady plume exiting the 14.75” magnet stinger, slowly insert stinger into the helium fill port and tighten brass fill cap as shown in Figure 8.

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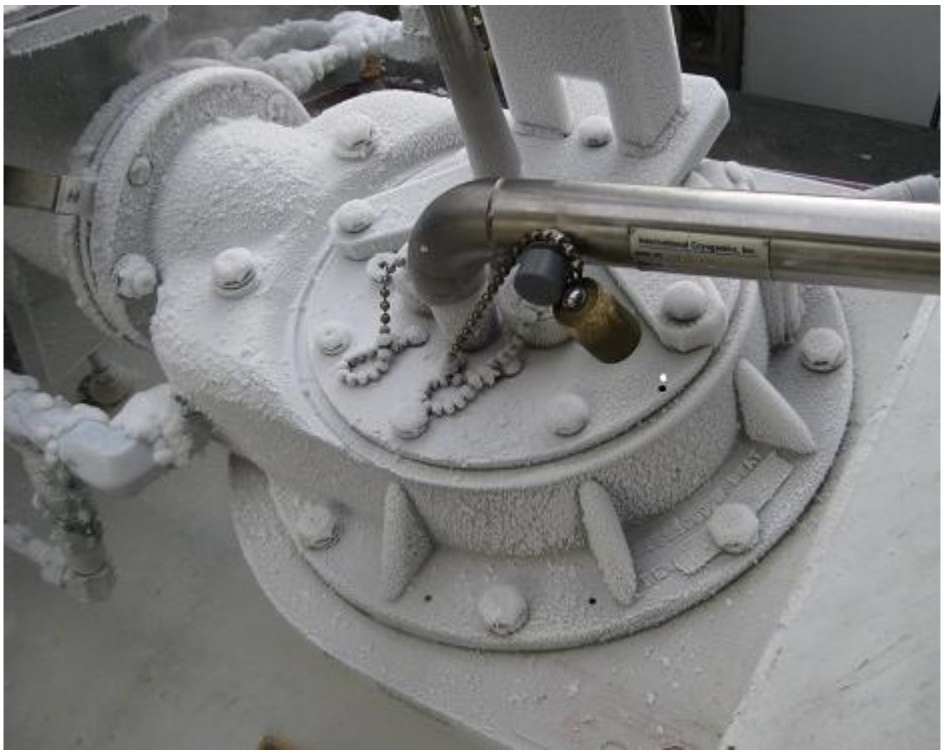

Figure 8

6.12 Fully open the Dewar end siphon valve and ensure the dewar is pressurized to 4.0 – 4.5 psi.

6.13 Monitor pressure and liquid level percentage change throughout the fill. If no percentage change is seen, check the dewar stinger and make sure it is pulled up 1/2 inch from bottom of dewar.

Caution: **A 250 Liter dewar will empty in approximately 25 to 30 minutes at 10 L/min. Always keep track of time and helium levels while transferring. Not keeping track of time and/or leaving a helium fill unattended may result in a quench.**

6.14 An empty dewar can be identified by one of the following indications:

1) Whistling sound can be heard from the dewar
2) The dewar pressure decreases (according to the dewar pressure gauge)
3) The magnet pressure increases.

Note:	**Helium level must not exceed 95% after transfer to avoid liquid from entering the recondensing system. Vent off excess liquid using V-2**

6.15 Once there is an indication that the dewar may be empty, or the helium level reaches 95% **stop the transfer** by closing the dewar syphon. **If additional dewars are required do not remove magnet stinger or close magnet vent (V2). Proceed to section 6.16**. If additional dewars are not required, close the magnet vent (V2). Remove the magnet stinger from the fill port (V1) replace brass cap and tighten (Make sure ½"O-Ring inside the Fill Port Cap is present, if not replace). <u>Proceed to section 6.21.</u>

6.16 If additional dewars are required, close valve on dewar stinger assembly of the empty dewar and discontinue the dewar pressurizing source.

6.17 Quickly disconnect the pressure building source from the empty dewar and connect to the full dewar.

6.18 Partially open the dewar stinger assembly valve on the full dewar until a liquid plume is observed**.**

6.19 Disconnect the helium transfer line from the dewar stinger of the empty dewar and when a plume is present, immediately connect it to the pluming dewar stinger.

6.20 Fully open dewar stinger assembly valve and verify the helium regulator pressure is at approximately 4.0 psig, continue the fill.

6.21 Using the 110v heat gun, warm up the fill port cap and retighten.

6.22 Leak check fill port cap using Snoop. If leaks are detected, remove fill port cap and inspect, make sure O-Ring inside cap is intact.

6.23 Replace fill port cap, tighten and leak check.

6.24 Turn the key on the Magnet Monitor counter clockwise and out of the "Fill" position as shown in Figure 9.

Figure 9

6.25 Record the final helium level and magnet pressure on Service Report.

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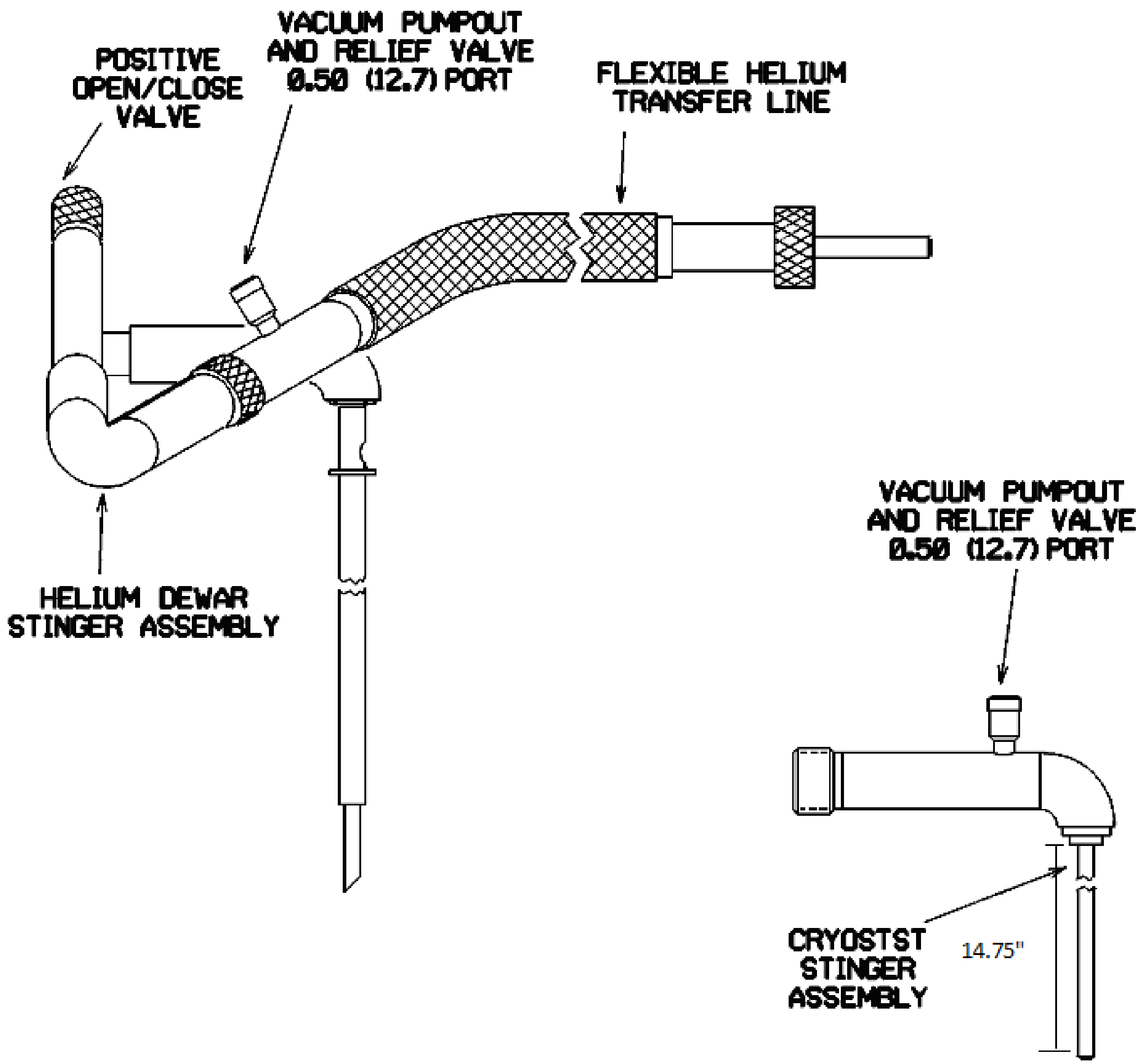

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Last Revision by: **This Revision By:** *C. Matthews*

APPROVAL	**MED**	**Service Operations Manager**	**Technical Operations Manager**	**Head MRI Service**
INITIALS	*MA*	*DG*	*CM*	*TF*
DATE	*1/13/12*	*1/11/12*	*1/11/12*	*1/12/12*

REVISIONS

Rev.	**Date**	**Changes**	
0	12/30/2011	S9495	First Issue.